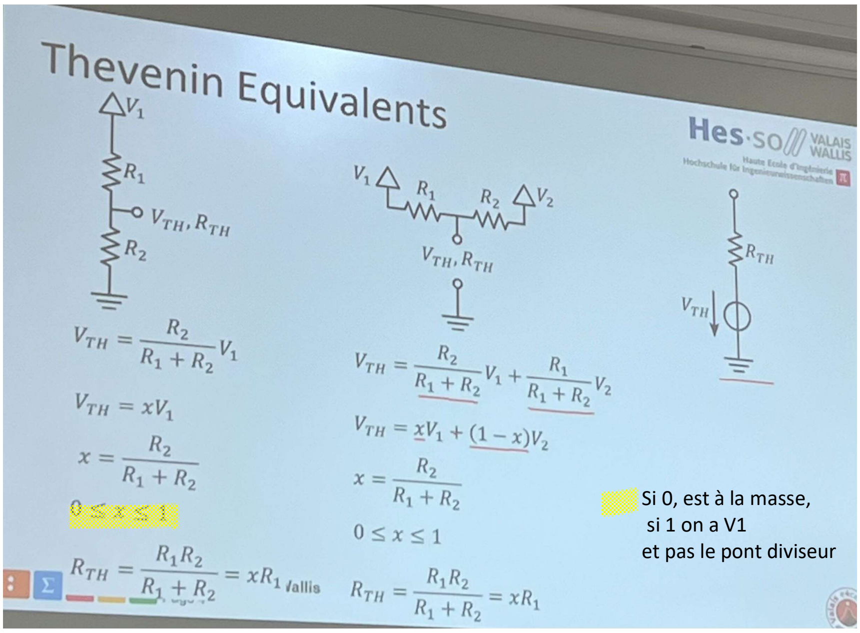
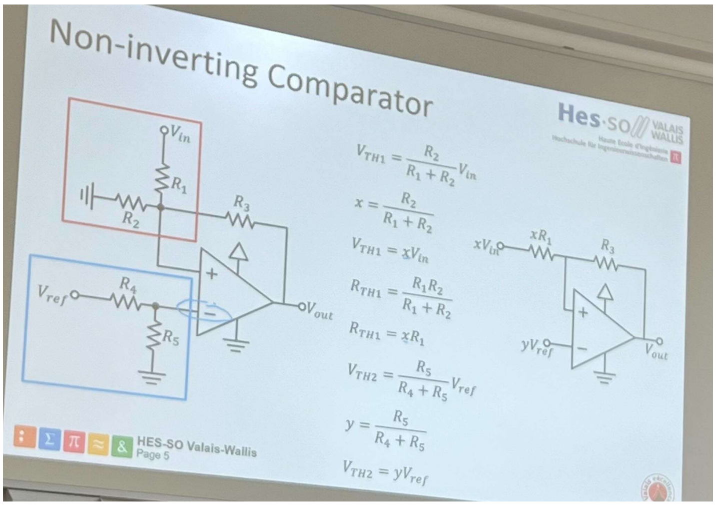
Labo 27.02 - Calculs

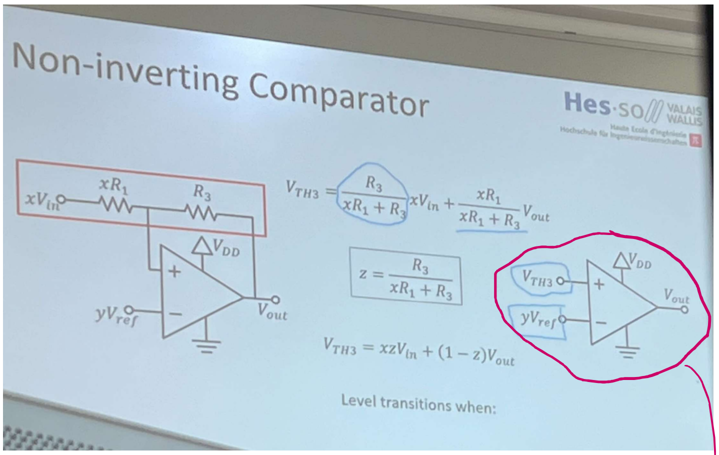
vendredi, 27 février 2026 12:59

Objectif trouver les résistances



Si 0, est à la masse,
si 1 on a V1
et pas le pont diviseur





$v_+ = V_-$

$yV_{ref} = xzV_{in} + (1-z)V_{out}$

$x = 1$ ou $y = 1$

passe à

$x = 1$ ou $y = 1$

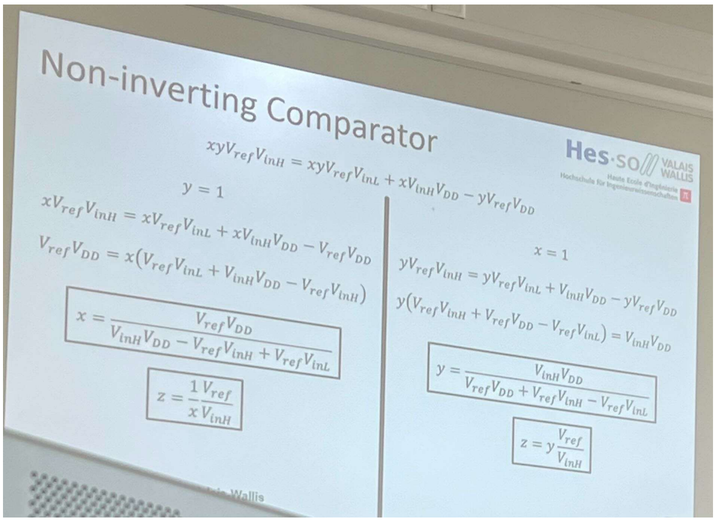
$$V_{out} = 0V \xrightarrow{\text{passe à}} V_{inH}$$

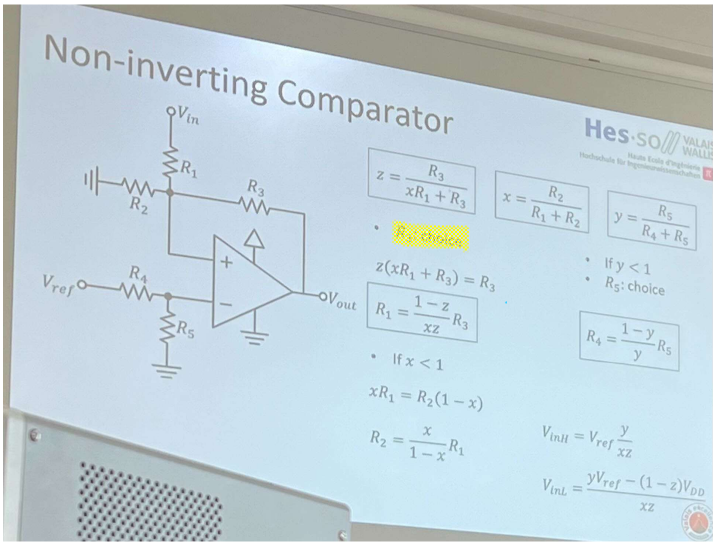
$$yV_{ref} = xzV_{inH}$$

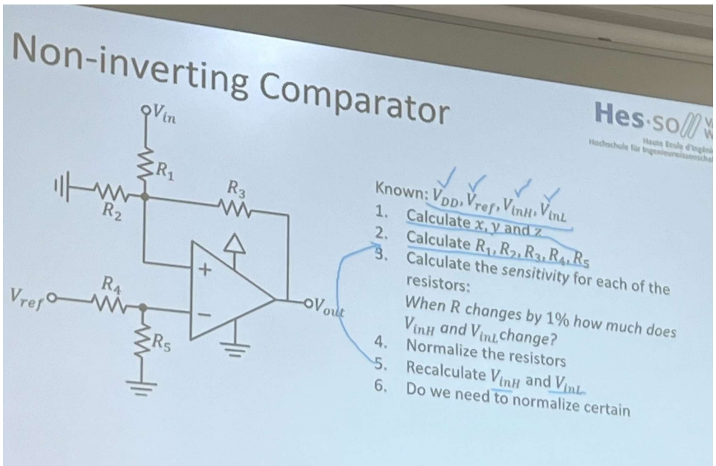
$$z = \frac{y}{x}\frac{V_{ref}}{V_{inH}}$$

$$V_{out} = V_{DD} \xrightarrow{\text{passe à}} V_{inL}$$

$$yV_{ref} = xzV_{inL} + (1-z)V_{DD}$$






Non-inverting Comparator
Hes·so
V_{in}
R_1
R_2
R_3
R_4
R_5
V_{ref}
V_{out}
Known: $V_{DD}, V_{ref}, V_{inH}, V_{inL}$
1. Calculate x, y and z
2. Calculate R_1, R_2, R_3, R_4, R_5
3. Calculate the *sensitivity* for each of the resistors:
When R changes by 1% how much does V_{inH} and V_{inL} change?
4. Normalize the resistors
5. Recalculate V_{inH} and V_{inL}
6. Do we need to normalize certain